

Target and stitched multiple-shooting estimate

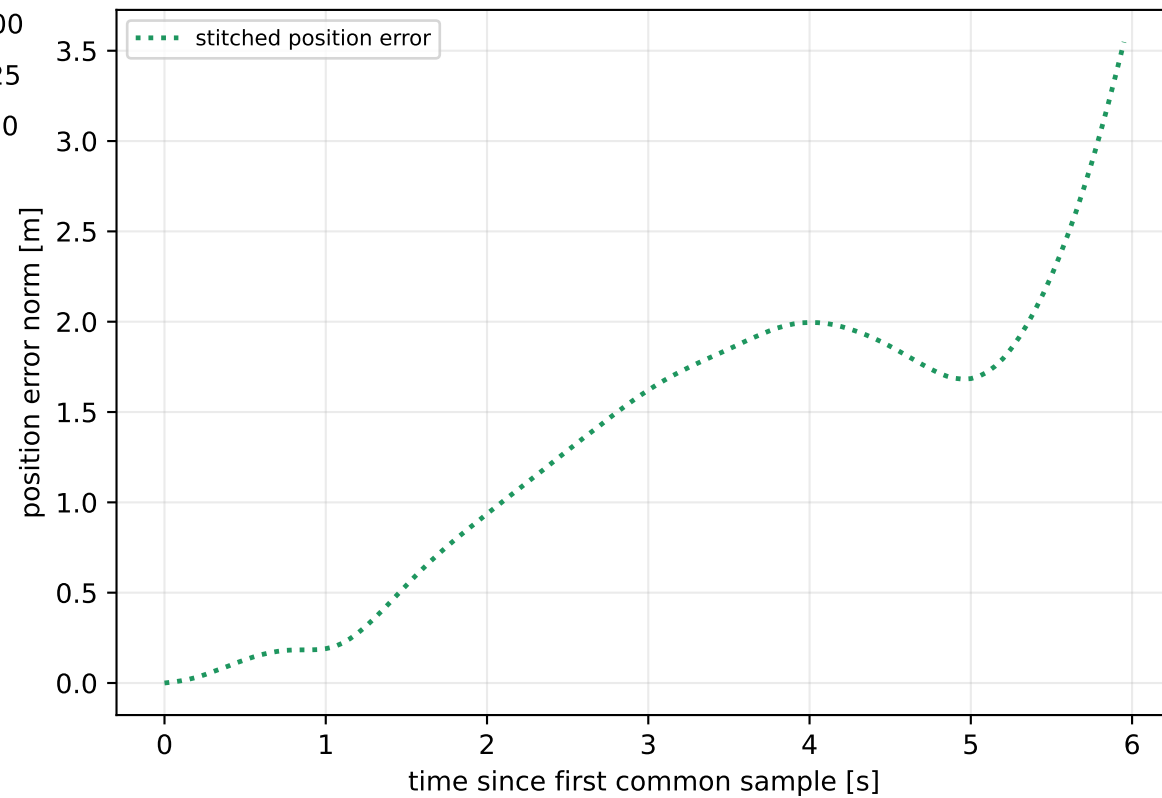
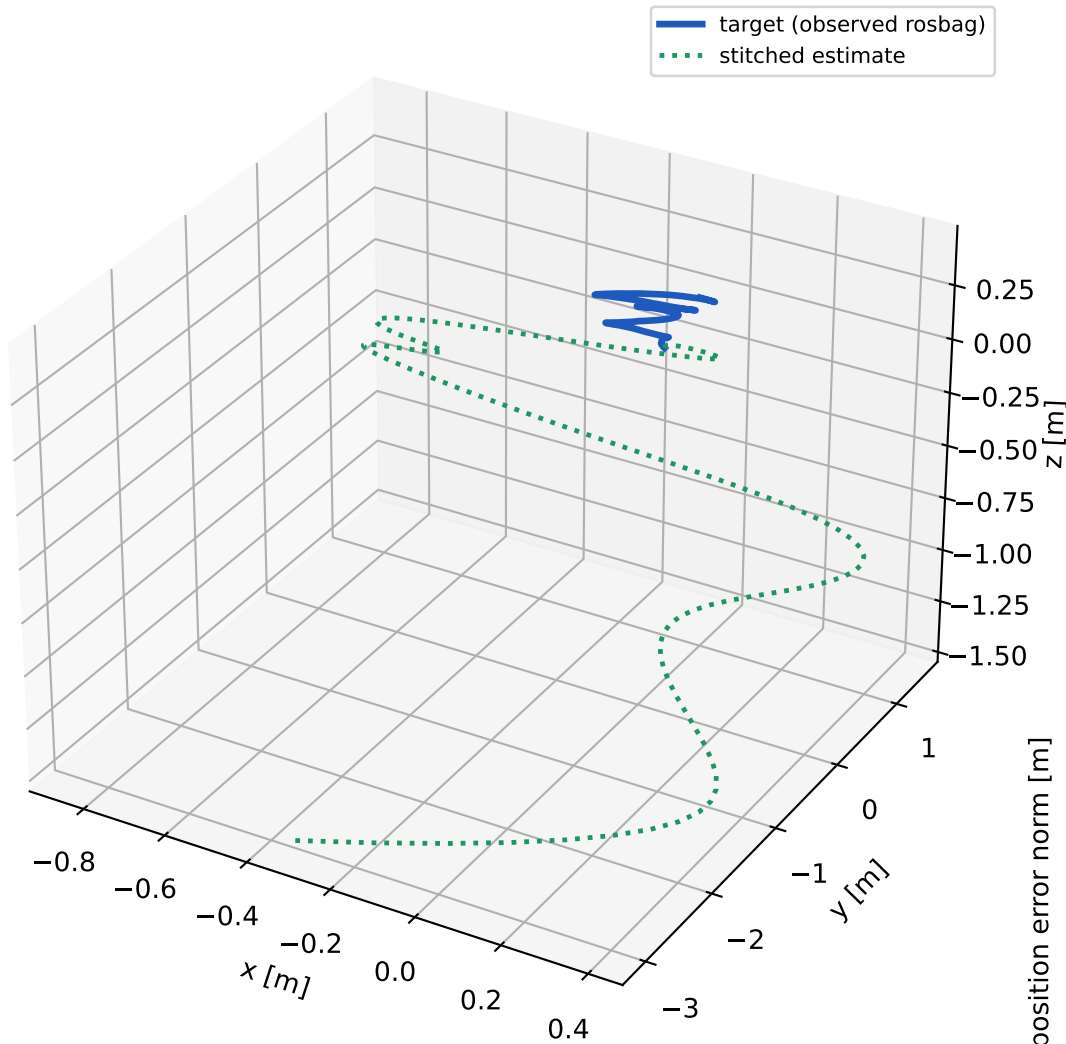
actual common support: 19.009–24.959 s
selected command delay: 0.126554 s

metric	stitched
position RMSE [m]	1.5657
orientation RMSE [deg]	20.948
velocity RMSE [m/s]	1.7115
angular velocity RMSE [rad/s]	0.70764
specific force RMSE [m/s ²]	1.2194

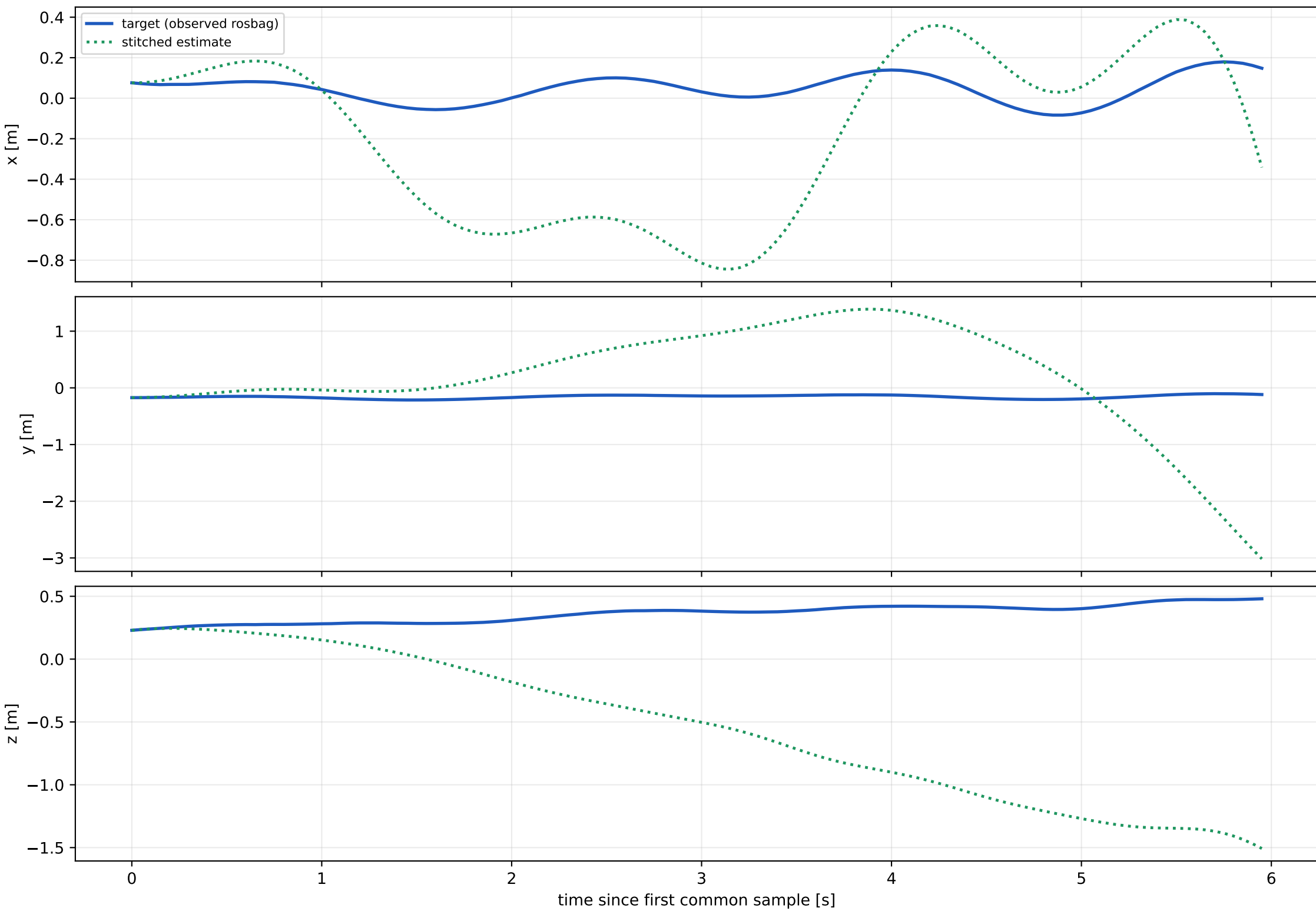
continuity max / tolerance:
4.441e-15 / 0.001 (converged)

Velocity and IMU metrics are diagnostic only;
the fit uses sensor position and orientation.

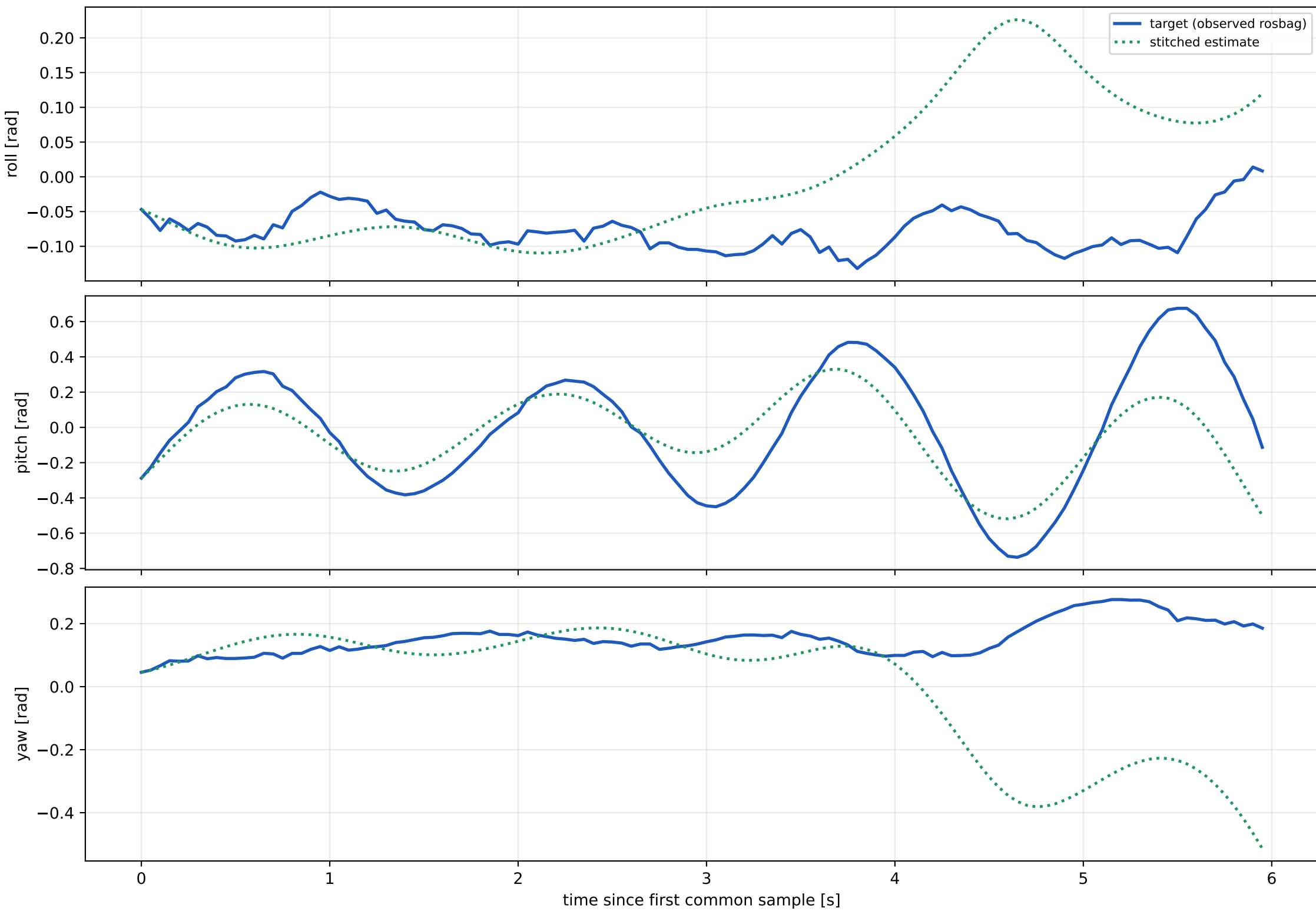
Stitched trajectory



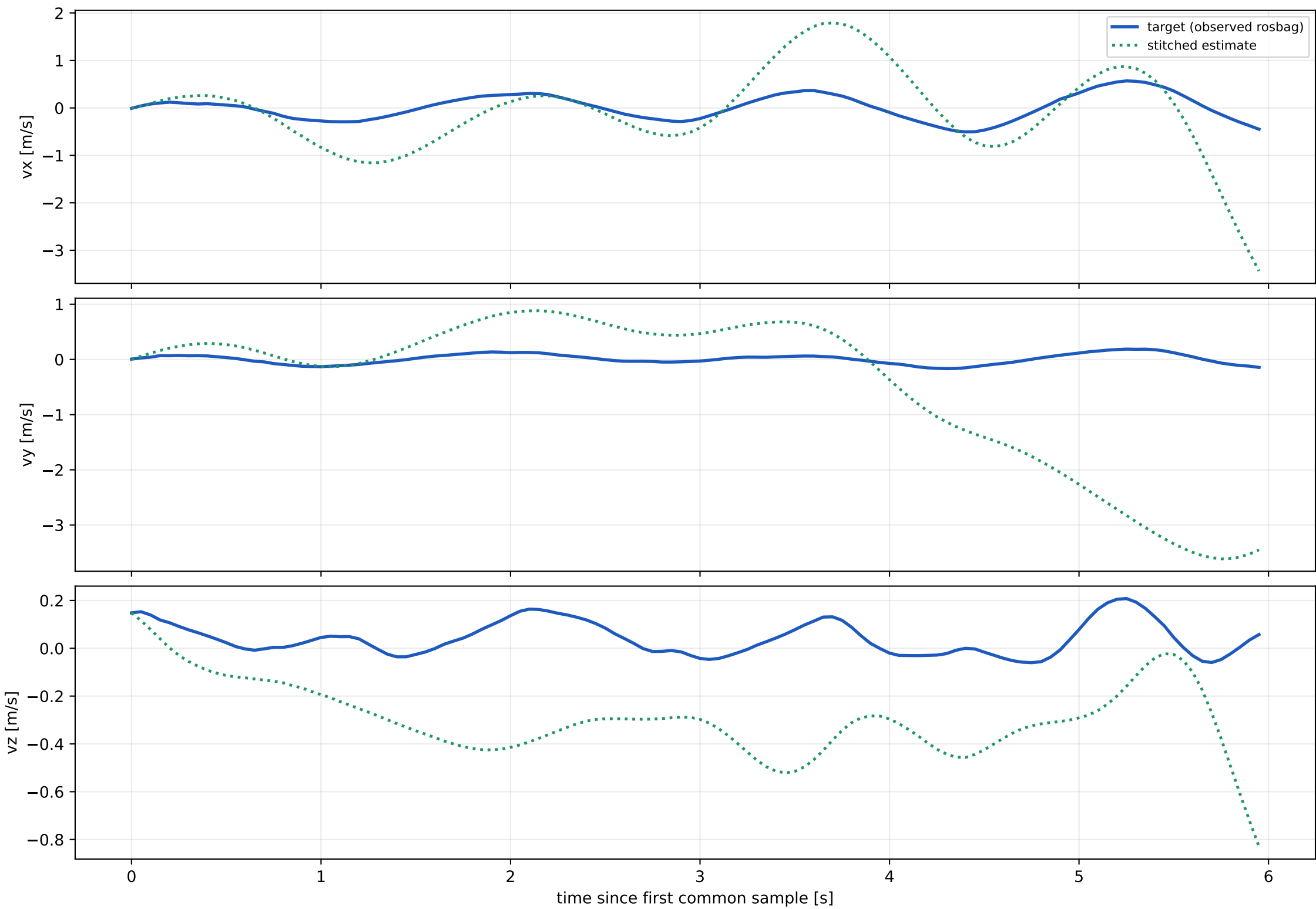
Sensor position



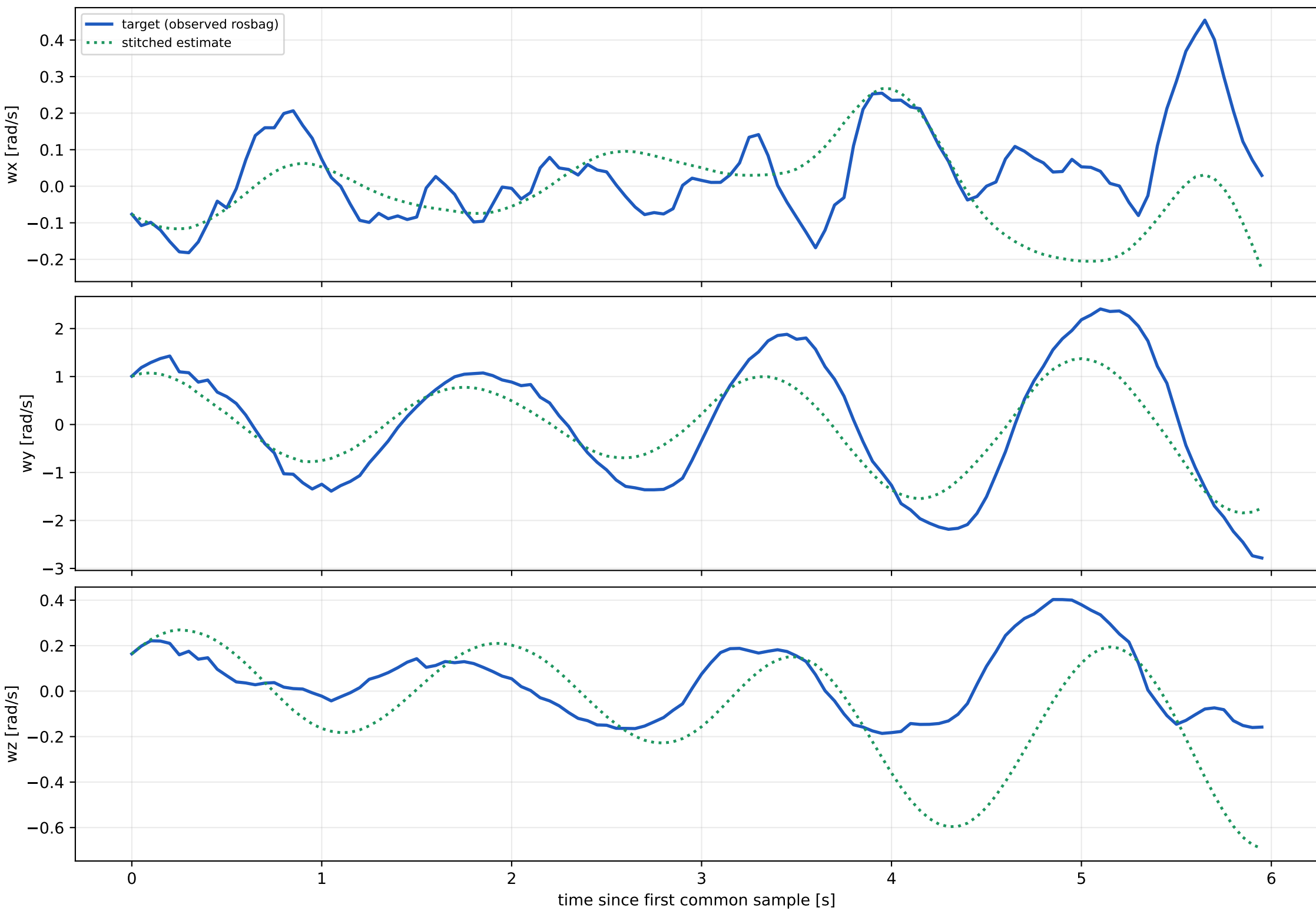
Sensor orientation (display only; fit uses SO(3) log)



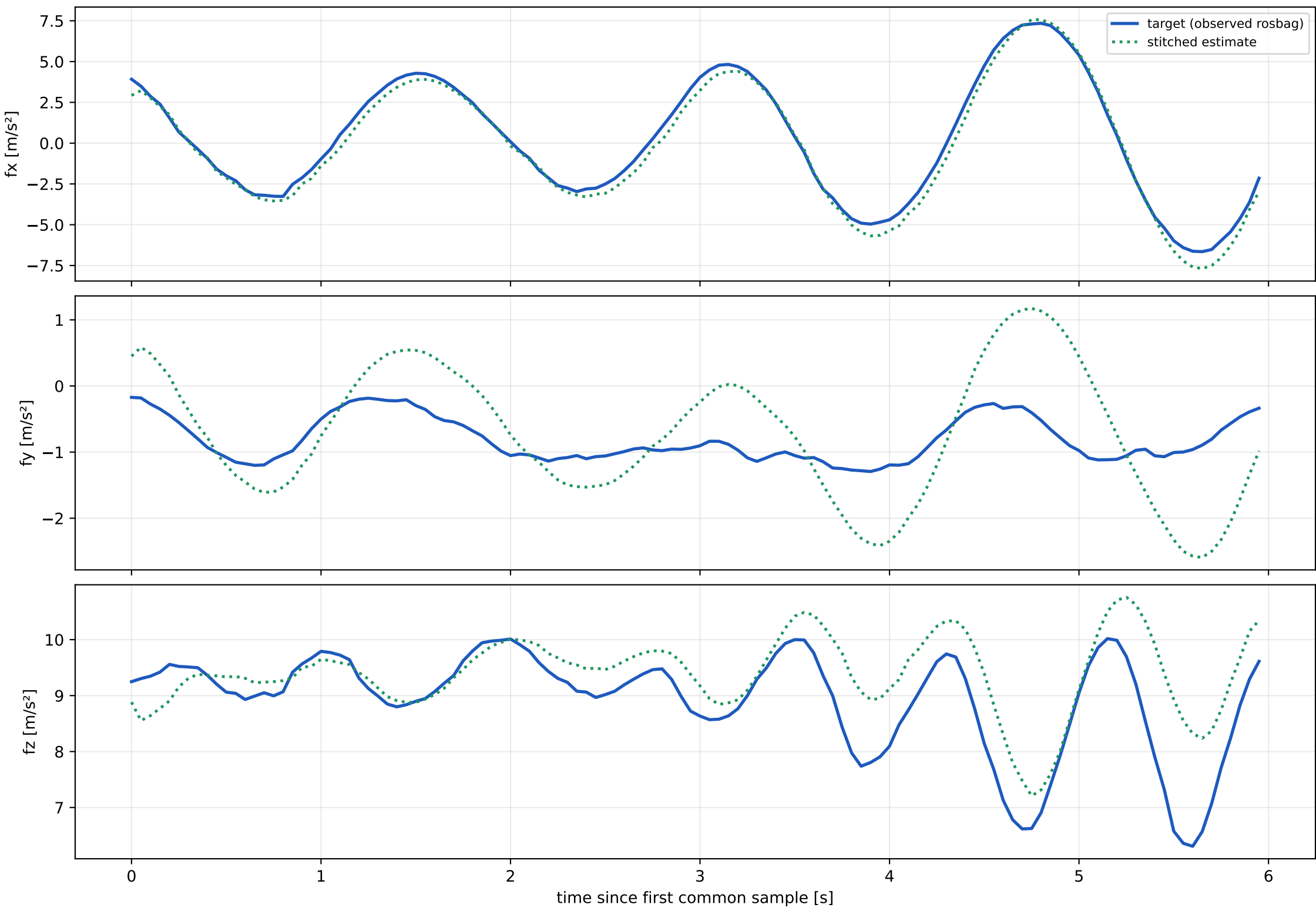
World-frame sensor velocity (diagnostic; excluded from fit)



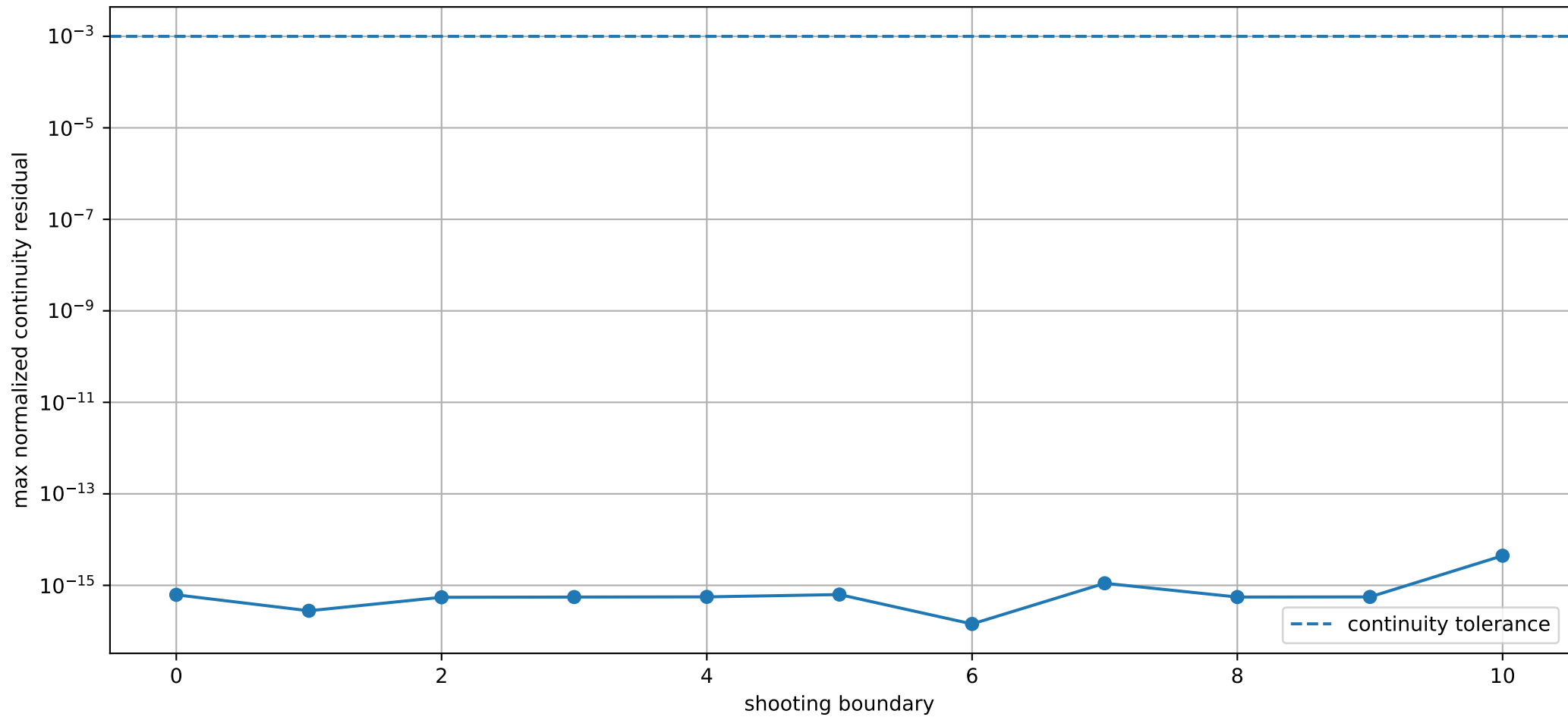
IMU angular velocity (diagnostic; excluded from fit)



IMU specific force (diagnostic; excluded from fit)



Shooting continuity diagnostic



Final estimated parameters

Selected multiple-shooting estimate

Decoded physical parameters

```
mass [kg] 3.02232298411
inertia [kg m^2]
[ 0.969421478511 0.109415232744 0.00269418769725]
[ 0.109415232744 0.259121283672 -0.0238195866122]
[ 0.00269418769725 -0.0238195866122 1.22640781476]
principal inertia [kg m^2] [0.242065895329, 0.985890660947, 1.22699402067]
CoG offset [m] [0.0611824169108, -0.0936319079332, -0.0104546517358]
rotor force effectiveness [1.01939436606, 2.27207200437, 0.702843051126, 0.614295487471]
command delay [s] 0.126553995291
```

Thirteen optimized smooth coordinates

```
log_mass_scale 0.250930812684
log_second_moment_cholesky_xx_scale 0.693465818191
log_second_moment_cholesky_yy_scale 1.32976612837
log_second_moment_cholesky_zz_scale 0.00131116299385
normalized_second_moment_cholesky_yx_offset -0.848128913489
normalized_second_moment_cholesky_zx_offset -0.070093160434
normalized_second_moment_cholesky_zy_offset 0.316546317295
cog_offset_x_m 0.0611824169108
cog_offset_y_m -0.0936319079332
cog_offset_z_m -0.0104546517358
force_effectiveness_contrast_1 -0.566734418026
force_effectiveness_contrast_2 0.630802486553
force_effectiveness_contrast_3 0.562661573802
```

Fit diagnostics

```
stitched inertia-radius loss [m^2] 1.23247730547
full-rollout inertia-radius loss [m^2] 1.23247730547
broad soft-prior cost 0.947972727871
position RMSE [m] 1.56567327932
orientation RMSE [deg] 20.94828793
velocity RMSE [m/s] 1.71154807377
angular velocity RMSE 0.707638240943 rad/s
specific force RMSE 1.21943352819 m/s^2
continuity max (normalized) 4.4408920985e-15
continuity tolerance 0.001
continuity converged True
```